	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the need for the removal of excretory products	To include the reasons for the deamination of amino acids and production of urea in liver cells (no details of the ornithine cycle are required).
(b)	the structure of the kidney as part of the excretory system	To include the gross structure of the kidney (cortex, medulla, renal pyramids, renal pelvis and ureter) and relative positions of the major blood vessels and organs associated with the kidney (renal artery and vein, bladder and urethra). **PAG2**
(c)	**(i)** the structure and function of the kidney nephron related to the processes of ultrafiltration and selective re-absorption which result in the production of urine	To include the use of microscopy, diagrams and photomicrographs to examine and identify structures. *M0.1, M0.2, M0.3, M1.8, M2.2* **PAG1** HSW8
	(ii) practical investigations into the biochemical composition of 'mock' urine, renal artery and renal vein plasma and filtrate	**PAG9** HSW3, HSW4, HSW5, HSW6
(d)	the role of the kidney in osmoregulation	To include the location and role of osmoreceptors, the secretion of ADH from the posterior pituitary, the action on ADH at the collecting ducts, the role of cyclic AMP in collecting duct cells and the insertion and removal of aquaporins into cell surface membranes.
(e)	the role of the kidney as an endocrine gland	To include an outline of the homeostatic function of erythropoietin (EPO) and renin (angiotensin).
(f)	the causes, diagnosis and consequence of kidney failure	To include causes (uncontrolled diabetes, kidney stones, blood pressure and bacterial infections), diagnosis (analysis of data from laboratory samples e.g. urine or blood composition) and consequences (renin and EPO changes and cardiovascular disease). *M0.1, M0.3, M2.2, M2.3, M2.4* **PAG9** HSW9, HSW10, HSW11, HSW12

(g)	the use of haemodialysis and peritoneal dialysis in the treatment of kidney failure	To include a consideration of the advantages and disadvantages of both forms. *M1.4* HSW9, HSW10, HSW11, HSW12
(h)	the use of transplant surgery in the treatment of kidney failure	To include practical issues involved in the use of donor organs for kidney transplants. HSW7
(i)	the future for transplant surgery.	To include the implications of therapeutic cloning and reproductive cloning and the use of stem cell technology. HSW7

2d. Prior learning and progression

This specification has been developed for learners who wish to continue with a study of biology at Level 3. The A level specification has been written to provide progression from GCSE Science, GCSE Additional Science, GCSE Further Additional Science, GCSE Biology or from AS Level Biology; achievement at a minimum of grade C (or equivalent) in these qualifications should be seen as the normal requisite for entry to A Level Biology. However, learners who have successfully taken other Level 2 qualifications in Science or Applied Science with appropriate biology content may also have acquired sufficient knowledge and understanding to begin the A Level Biology course.

There is no formal requirement for prior knowledge of biology for entry onto this qualification. Other learners without formal qualifications may have acquired sufficient knowledge of biology to enable progression onto the course.

Some learners may wish to follow a biology course for only one year as an AS, in order to broaden their curriculum, and to develop their interest and understanding of different areas of the subject. Others may follow a co-teachable route, completing the one–year AS course and/or then moving to the two–year A level.

The A Level Biology course will prepare learners for progression to undergraduate study, enabling them to enter a range of academic and vocational careers in Biological Sciences, Medicine and Biomedical Sciences, Veterinary Science, Agriculture and related sectors. For learners wishing to follow an apprenticeship route or those seeking direct entry into biological science careers, this A level provides a strong background and progression pathway.

There are a number of Science specifications at OCR. Find out more at www.ocr.org.uk